

Marina C. Kapitanov

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Motivated and creative biomedical engineering student seeking a research opportunity that builds on communication and technical skills in diving physiology and medicine.

EDUCATION

North Carolina State University (NC State), Raleigh, NC

BS, Biomedical Engineering, minors in Spanish and German

GPA: 3.529

Expected Graduation: May 2025

Nagoya University, Nagoya, Japan

Nagoya University Program for Academic Exchange

September 2023 - August 2024

Heidelberg University, Heidelberg, Germany

UNC-BW Consortium Exchange

March 2022 - August 2022

WORK EXPERIENCE

Undergraduate Research Assistant, Divers Alert Network,

Divers Alert Network, Durham, NC

January 2023 - Present

Analyzing salivary stress and inflammation analytes and venous gas emboli scores from cardiac ultrasounds amongst different divers diving the same standardized profile to study variation and possible causes of decompression stress and sickness

Guided Independent Research Project, Biomechanics Lab,

Nagoya University, Nagoya, Japan

April 2024 - August 2024

Investigated effects of hyperbaric pressure on mouse osteoblast-like cells to better understand osteogenesis misbalance mechanisms of dysbaric osteonecrosis

Undergraduate Research Assistant REdesign Lab,

North Carolina State University, Raleigh, NC

October 2022 - May 2023

Helped develop a micro-tissue platform to study engineered models of muscle and cardiac tissue. Manufactured well plates, seeded tissues, and evaluated tissue contraction within the plate using image analysis

Undergraduate Research Assistant, Transmembrane Signaling Laboratory

Bulgarian Academy of Sciences, Institute for Biophysics and Biomedical Engineering, Sofia, Bulgaria

January 2022 - March 2022

Synthesized new multi-chromophore chemosensory systems for cell marking and diagnostics and investigated pathophysiological relationships between phospholipid and Akt pathways in an *in vivo* Alzheimer's model

Undergraduate Research Assistant, Orthopaedic Mechanobiology Lab

North Carolina State University, Raleigh, NC

May 2021 - August 2021

Utilized and edited a custom program to develop the beginning stages of two 3D finite element models of brachial plexus birth injury in a rat model

AWARDS & FELLOWSHIPS

NC State Park Scholar

August 2020 - Present

NC Japan Center Harry C. Kelly Memorial Fund Awardee

April 2024 - August 2024

Abrams Research Scholar, Joint Department of Biomedical Engineering NC State & UNC

Oct 2022 - May 2024

NC State Women and Minority Engineering Summer Research Program

May 2021 - August 2021

PROFESSIONAL ACTIVITIES & MEMBERSHIP

Undersea and Hyperbaric Medicine Society Member

Summer 2024 - Present

PackAbroad Ambassador

Fall 2022 - Present

Engineers Without Borders USA: NC State Student Chapter Grant Writing Team

Fall 2020 - Present

University Scholars Program

Fall 2020 - Present

POSTERS AND PUBLICATIONS

Brenner, R. J., Currens, J. B., Eltz, K., Coombs, M., Andersen, M., **Kapitanov, M. C.**, Linger, A. S., Grant, D., Leypoldt, J., Dong, G. Z., Harris, C. J., Dugrenot, E., Thom, S. R., Papadopoulou, V., & Tillmans, F. (poster, 2024). Midpoint assessment: a prospective trial on intra- and inter-subject variability in venous gas emboli and other biomarkers after repeated dives. Undersea Hyperbaric Medical Society Annual Scientific Meeting (New Orleans, Louisiana).

Berger, M. B., Cohen, D. J., Bosh, K. B., **Kapitanov, M.**, Slosar, P. J., Levit, M. M., Gallagher, M., Rawlinson, J. J., Schwartz, Z., & Boyan, B. D. (2023). Bone marrow stromal cells generate an osteoinductive microenvironment when cultured on titanium-aluminum-vanadium substrates with biomimetic multiscale surface roughness. *Biomedical materials* (Bristol, England), 18(3), 035001.

SKILLS

Bulgarian, Spanish, German, low-intermediate Japanese; cell culture, image analysis, fieldwork; open water SCUBA